

ElectroMart — E-Commerce Platform for Electrical Shop

Project Blueprint v1.0

Stack: Angular (Frontend) + Spring Boot (Backend, Monolithic) + MySQL/PostgreSQL

Payments: Razorpay + Cash on Delivery (COD) **Media:** Cloudinary **Communication:** Email (OTP, order confirmation, invoices) via SMTP/SendGrid

1. Tech Stack (Detailed)

Layer	Technology
Frontend	Angular 17+ (Standalone Components), TypeScript, RxJS, Angular Material / Tailwind CSS
Backend	Spring Boot 3.x, Spring MVC, Spring Security, Spring Data JPA
Database	MySQL / PostgreSQL
Auth	JWT (Access + Refresh Token), Spring Security
Caching	Redis (for cart, OTP, product cache) — optional but recommended for "premium" feel
Media Storage	Cloudinary (product images, banners, category icons)
Payment Gateway	Razorpay (Orders API + Webhooks) + COD
Email Service	JavaMailSender + SMTP (or SendGrid/Amazon SES) — OTP, order confirmation, invoice PDF
Search	MySQL full-text / Elasticsearch (optional upgrade)
Deployment	Docker + Nginx + AWS/GCP/DigitalOcean or Hostinger VPS
CI/CD	GitHub Actions (optional, adds "professional" value)
Monitoring	Spring Boot Actuator + basic logging (ELK optional)

Architecture: Monolithic Spring Boot backend exposing REST APIs, consumed by Angular SPA. Clean layered structure: Controller → Service → Repository → Entity, with DTOs, Global Exception Handling, and Interceptors/Filters for JWT.

2. Theme & UI/UX Direction

Since it's an **electrical shop**, the theme should feel technical, trustworthy, and energetic — not generic.

- **Color palette:** Deep navy/charcoal base + electric yellow/amber accent (think "voltage" — safety-yellow highlights on dark/white backgrounds). Alternative: Blue + Orange (trust + energy).
 - **Typography:** Clean sans-serif (Inter, Poppins, or Roboto) — modern, readable, no clutter.
 - **Visual motifs:** Subtle circuit-line patterns, bolt/plug icons, badge icons for "Genuine Product", "ISI Marked", "Warranty Included".
 - **Layout style:** Card-based product grid, sticky header with mega-menu (categories like Wiring, Switches, MCBs, Lighting, Fans, Tools, Motors), large banner carousel for offers.
 - **Mobile-first & responsive** — most electrical shop customers (electricians, contractors) will browse on mobile.
 - Dark-mode toggle (optional nice-to-have, adds "professional SaaS" feel).
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3. Feature List (Go List-Wise From Here)

A. Customer-Facing Website

3.1 Authentication & Account

- Signup/Login (Email + Password)
- Email OTP verification on signup
- Forgot password via OTP/email link
- Login with Google (OAuth2) — optional premium add-on
- JWT-based session, "Remember me"
- My Account dashboard: Orders, Addresses, Wishlist, Wallet/Refunds, Profile edit

3.2 Product Catalog

- Category & sub-category navigation (e.g., Wires & Cables → Sub: Copper, PVC)
- Product listing with filters: Brand, Price range, Rating, Availability, Specifications (Voltage, Wattage, Material etc.)
- Sorting: Price, Popularity, Newest, Rating
- Product detail page: Multiple images (zoom/gallery via Cloudinary), specs table, description, brand, warranty info, stock status
- Related/Similar products, "Frequently bought together"
- Product Q&A / Ask a question
- Ratings & Reviews (with image upload, verified purchase badge)

- Bulk/Wholesale pricing tiers (great feature for electrical B2B customers — electricians buying in bulk)

3.3 Search

- Autocomplete/typeahead search
- Search by product name, brand, category
- "No result" smart suggestions

3.4 Cart & Checkout

- Add to cart, update quantity, save for later
- Wishlist
- Guest checkout (optional) + Login checkout
- Address management (multiple addresses, default address)
- Pincode-based delivery check
- Coupon/Promo code engine
- Order summary with tax (GST) breakdown
- Payment: Razorpay (UPI/Card/Netbanking/Wallet) + COD option
- COD limit rules (e.g., disable COD above ₹X or for certain pincodes)

3.5 Order Management (Customer side)

- Order confirmation email with invoice PDF
- Order tracking with status timeline (Placed → Confirmed → Packed → Shipped → Delivered)
- Order history, reorder button
- Cancel order / Return-Replace request
- Downloadable GST invoice

3.6 Engagement Features

- Newsletter subscription
- Notification center (order updates, offers)
- Product stock alert ("Notify me when available")
- Recently viewed products
- Compare products (side-by-side specs — useful for electrical items like MCBs, fans)

3.7 Static/Trust Pages

- About Us, Contact Us (with map/store location), FAQs
- Shipping & Return Policy, Terms, Privacy Policy
- Warranty/Brand assurance page

- Blog/Guides section (e.g., "How to choose the right MCB") — great for SEO
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B. Admin Panel

3.8 Dashboard

- Sales overview, revenue graphs (daily/monthly), best-selling products, low-stock alerts

3.9 Catalog Management

- CRUD for Products, Categories, Sub-categories, Brands
- Bulk product upload via CSV/Excel
- Image upload directly to Cloudinary from admin panel
- Inventory/stock management with low-stock threshold alerts

3.10 Order Management

- View/manage all orders, update order status
- Generate/download invoices
- Refund management (Razorpay refund API integration)
- Return/replacement handling

3.11 User Management

- View customers, block/unblock
- Role-based access (Super Admin, Manager, Staff) — Spring Security roles

3.12 Marketing

- Coupon/discount management
- Banner/carousel management (Cloudinary upload)
- Featured products / "Deal of the day" configuration

3.13 Reports

- Sales report, inventory report, customer report (exportable to Excel/PDF)
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C. Cross-Cutting / "Professional" Features (These Justify the Price Tag)

- **GST-compliant invoicing** (auto tax calculation, HSN codes)
- **SEO optimization:** Angular Universal (SSR) or prerendering for product pages, meta tags, sitemap.xml, robots.txt
- **Performance:** Lazy loading modules, image lazy-loading, Cloudinary auto-optimized image delivery (WebP, responsive sizes), API response caching

- **Security:** Spring Security + JWT, password hashing (BCrypt), rate limiting on OTP/login, HTTPS, input validation, CORS config, SQL injection/XSS protection
 - **Razorpay Webhooks** for reliable payment status updates (not just client-side callback)
 - **Email templates:** HTML templates for OTP, order confirmation, shipping updates, invoice attachment (PDF via iText/OpenPDF)
 - **Audit logging** for admin actions
 - **Responsive PWA support** (installable, works offline for browsing) — strong selling point
 - **Google Analytics / Meta Pixel integration** ready
 - **Abandoned cart email reminder** (nice premium touch)
 - **Multi-image drag-drop uploader** in admin (Cloudinary widget)
 - **Automated backups** (DB backup strategy documented for client)
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4. Suggested Database Modules (High-Level)

User, Role, Address, Category, SubCategory, Brand, Product, ProductImage, ProductVariant, Review, Cart, CartItem, Wishlist, Order, OrderItem, Payment, Coupon, Refund, Notification, OtpVerification, Banner
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5. Suggested Development Phases

1. **Phase 1 — Foundation:** Project setup (Angular + Spring Boot skeleton), DB schema, Auth + OTP + JWT
 2. **Phase 2 — Catalog:** Category/Product CRUD, Cloudinary integration, listing + filters + search
 3. **Phase 3 — Cart & Checkout:** Cart, address, coupon engine, Razorpay + COD integration
 4. **Phase 4 — Orders & Admin:** Order lifecycle, admin dashboard, invoice/email system
 5. **Phase 5 — Polish:** Reviews, wishlist, compare, SEO/SSR, PWA, performance tuning
 6. **Phase 6 — Deployment:** Dockerize, deploy, SSL, monitoring, handover docs
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Next Step

Tell me which list you want to start with — e.g., **DB schema design, module-by-module API design**, or **Angular folder/module structure** — and we'll go through it in detail, list-wise.